

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING ASSESSMENT BASED PROJECT

(BCS602)

TITLE: Phish Protect AI: An AI-Driven Framework for Real-Time Phishing Threat Intelligence

ABSTRACT

This study addresses the growing risk of phishing attacks resulting from increased online transactions, where traditional blacklist and heuristic methods often fail to detect modern, evolving threats. An automated phishing website detection system is proposed using supervised machine learning models specifically Random Forest and XG Boost to classify websites based on extracted URL-based and content-based features. Experimental evaluation shows that both models achieve high accuracy with minimal false positives and false negatives, with XG Boost demonstrating slightly better performance. The results confirm that ensemble-based techniques can effectively support scalable, adaptive, and real-time phishing detection solutions. Future work will incorporate explainable AI to enhance model transparency and user trust.

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